Bone remodeling is the continuous, microscopic process by which bone tissue throughout the body is resorbed and redeposited. Proper regulation of bone remodeling is necessary to maintain bone strength and prevent the onset of fragile bone conditions such as osteoporosis. Patients with such conditions have increased fracture risk and generally exhibit decreased osteoblast (bone-depositing cell) activity, leading to reduced bone density and weakening of the bone matrix. However, some cases can occur due to overactive osteoclasts (multinucleated cells that mediate bone resorption). Osteoclasts resorb bone by secreting proteolytic enzymes and acid into the adjacent extracellular space. These enzymes degrade the organic matrix of bone, and the acid causes the dissolution and release of the mineral components of bone such as calcium, potassium, and magnesium.

Osteoclast differentiation and activation is thought to be related to the RANKL/RANK signaling pathway. RANKL, expressed as a membrane-bound or secreted protein, serves as the ligand for RANK, a transmembrane receptor. In bone tissue, RANK is expressed by precursors of mature osteoclasts, also known as osteoclast progenitor cells (OPCs). RANKL can be expressed by osteoblasts adjacent to OPCs, and the binding of RANKL to RANK in these neighboring cells causes downstream effects through recruitment of TRAF6, an intracellular protein. RANK is involved in multiple downstream signaling cascades, and four of these directly mediate the development of mature osteoclasts from OPCs. In one of the four pathways, the transcription factor NF-κB is activated, which leads to increased expression of the proto-oncogene c-Fos. Subsequently, c-Fos interacts with another transcription factor known as NFATc to promote the expression of genes related to osteoclast-specific development and function in OPCs (Figure 1).

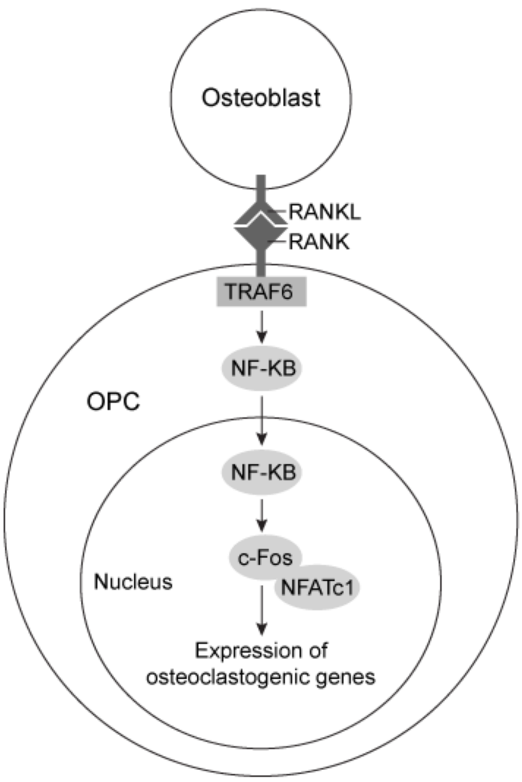


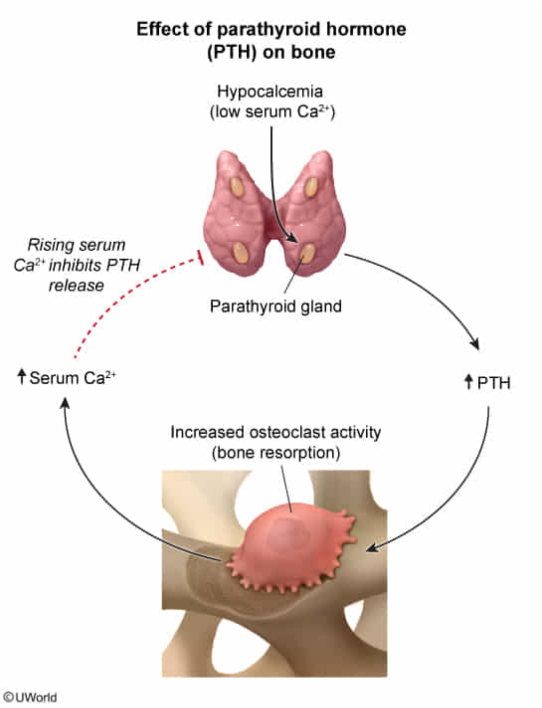
Figure 1 RANKL/RANK signaling induces differentiation of OPCs into osteoclasts

The information in the passage suggests that increased activity of the RANKL/RANK signaling pathway is most likely due to signaling involving which of the following hormones?

- ✔ ◉ A. Parathyroid hormone [49%]
- ○ B. Calcitonin [40%]
- ○ C. Growth hormone [9%]
- ○ D. Glucagon

Correct
48% answered correctly

Explanation:



Endocrine glands secrete hormones that travel through the bloodstream and bind target receptors in specific tissues, evoking responses that promote the maintenance of physiological homeostasis. For example, parathyroid hormone (PTH) and calcitonin are hormones that maintain **calcium homeostasis** by regulating calcium absorption and **bone remodeling**.

Specifically, a low blood calcium level stimulates the parathyroid glands to secrete **PTH**, which then promotes several processes to elevate the blood calcium level. One of these processes involves a PTH-induced **increase in** the **activity of osteoclasts**, which can actively break down (resorb) bone, thereby leading to release of calcium from bone into the bloodstream.

According to the passage, the activation of the RANK receptor by RANKL stimulates the differentiation of OPCs into mature osteoclasts. Because PTH induces **increased activity of osteoclasts**, it is likely associated with **increased activity of the RANKL/RANK signaling pathway** and subsequent stimulation of osteoclast maturation and activity.

(Choice B) Calcitonin is secreted by the thyroid gland in response to *high blood calcium levels* and *opposes* PTH activity by *decreasing* the activity of osteoclasts. Therefore, calcitonin would likely be associated with decreased (*not* increased) RANKL/RANK activity in osteoclasts.

(Choice C) The anterior pituitary gland secretes growth hormone (GH) to induce tissue growth, including bone development. GH can *stimulate* osteoblast activity to increase bone matrix deposition. Because GH is a key promoter of bone *growth* in children and increases the rate of cell turnover in adults, it is *not* likely associated with *increased breakdown* of bone by osteoclasts (ie, *increased* RANKL/RANK activity).

(Choice D) Glucagon, a hormone secreted by the pancreas, is a key regulator of blood glucose (*not* bone remodeling). In the setting of low blood glucose, glucagon stimulates the breakdown of glycogen in the liver to increase the blood glucose level.

Educational objective:
To regulate calcium homeostasis, parathyroid hormone (PTH) and calcitonin act antagonistically to each other. PTH is secreted in response to a low blood calcium level and stimulates bone resorption by osteoclasts, causing an increase in blood calcium. In contrast, calcitonin is secreted in response to a high blood calcium level and decreases osteoclast activity, ultimately decreasing blood calcium.

Time spent:0 sec	**QID**:401904
Subject: Biology	**System:** Musculoskeletal System